



Research Article

Designing flame retardants for epoxy resins with generative AI for preventing battery thermal runaway propagation



Pooya Jafari^{a,1}, Guofeng Ye^{b,1}, Junling Wang^{c,1}, Tao Chu^a, Boyou Hou^a, Di Wu^d, Siqi Huo^{a,*}, Hao Wang^e, Ziqi Sun^{f,*}, Pingan Song^{a,*}

^a Centre for Future Materials, School of Science, Engineering and Digital Technologies, University of Southern Queensland, Springfield 4300, Australia

^b Hubei Engineering Technology Research Centre of Optoelectronic and New Energy Materials, School of Materials Science & Engineering, Wuhan Institute of Technology, Wuhan 430205, China

^c Jiangsu Key Laboratory of Hazardous Chemicals Safety and Control, College of Safety Science and Engineering, Nanjing Tech University, Nanjing 211816, China

^d School of Computing, Engineering and Mathematical Sciences, La Trobe University, Victoria 3086, Australia

^e School of Engineering, Swinburne University of Technology, Hawthorn, VIC 3122, Australia

^f School of Chemistry and Physics, Queensland University of Technology, Brisbane, QLD 4000, Australia

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ABSTRACT

Artificial intelligence (AI) has demonstrated great potential for discovering new flame retardants (FRs) for polymers. While existing AI can describe flame retardancy of FRs, it has been unable either to generate new FR molecules with desired performances or predict their impacts on mechanical strength and glass transition temperature (T_g) of polymer matrices. To fill this knowledge gap, we, herein, propose a generative artificial intelligence (GAI)-based de novo molecular design approach (GAI4FR) to generate new FR molecules for commercially important epoxy (EP) resins. Also, a descriptive AI model is trained using existing works to predict impacts of AI-generated FR molecules on flame retardancy, tensile strength (σ_t), and T_g of EP, enabling the identification of three FR molecules with better overall performances. The as-identified FR molecules are then synthesized, and their predicted performances are well-validated. We further demonstrate the application of as-prepared flame-retardant EP-coated wood sheets as heat shields for preventing thermal runaway of lithium-ion battery (LIB) packs. The coated wood shows equally desirable thermal protection for LIB packs to commercial counterparts. This work offers a groundbreaking GAI4FR framework for creating next-generation high-performance flame retardants for various flammable polymers and opens the door to discovering many other functional materials.

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1. Introduction

EP resins, as a kind of industrially commonly-used polymeric material, feature high mechanical strength and exceptional resistance to heat and chemicals while being inexpensive and easy to manufacture [1–6]. These properties make EP indispensable across various industries, e.g., automobile, aerospace, electronics & electrical devices and batteries [7,8]. However, EPs are highly flammable, significantly limiting their practical applications in sections with demanding high safety requirements [9]. To address this issue,

many phosphorus-containing flame retardants (FRs) have been developed for EP because of their high efficiency and low toxicity [10–15]. Unfortunately, the existing discovery approaches of FRs heavily rely on researchers' experience and costly trial-and-error experiments [16,17], and often lead to flame retardants with limited efficiency and unwanted overall performances.

The emergence of materials databases, data science, and artificial intelligence (AI) have enabled material design to advance to a new era [18], where data-driven approaches are currently used to tackle the challenges of inverse materials design [19]. These techniques have been regarded as the “fourth pillar” of scientific research, alongside theoretical methods, computer simulations, and experimental techniques [20]. Many studies have investigated the de novo design of functional compounds using data-driven methods, and successful examples include material informatics [21], polymer genome [22], and polymer informatics [23].

* Corresponding authors.

E-mail addresses: Siqi.Huo@unisq.edu.au, sqhuo@hotmail.com (S. Huo), ziqi.sun@qut.edu.au (Z. Sun), pingan.song@usq.edu.au (P. Song).

¹ These authors contributed equally to this work.

Recently, machine learning (ML), a subdivision of AI, has been regarded as an effective tool for inverse molecular design [24]. ML-based methods have demonstrated great potential for addressing challenges in materials design and accelerating new materials discovery [25]. To date, ML has been employed to discover new FRs, but has been limited to uncovering the correlations between characteristic variables (e.g., addition amount and functional group) and flame retardancy, e.g., limiting oxygen index (LOI), UL-94 rating and peak heat release rate (PHRR) [26,27]. LOI, UL-94, and PHRR are critical indicators for evaluating the flame retardancy and self-extinguishing performance of polymer materials. For instance, Nguyen et al. [28] utilized an artificial neural network model to predict the HRR of phenolic resin loaded with ammonium polyphosphate (APP), sepiolite, and glass fibre. Zhang et al. [29] constructed a database of over 800 types of nanocomposites and investigated the effects of nanofillers and intumescent flame retardants on the physical properties of the nanocomposites. Chen et al. [30] reported an ML-based adaptive system capable of effectively guiding the selection and prediction of new flame retardant combinations. In another study [31], an ML model was developed to correlate the addition amounts and structure of FRs and the LOI of the final materials. In addition, Chen et al. [32] applied sure independence screening and a sparsifying operator approach to investigate the impacts of flame retardant components on LOI values. Despite these encouraging advancements, to date, existing ML models have been unable to generate new highly efficient FR molecules for specific polymers. Also, current ML regression models cannot effectively predict the impacts of FRs on the mechanical properties and glass transition temperature (T_g) of polymer matrices.

To bridge this knowledge gap, we propose a GAI4FR de novo molecular design framework (GAI4FR, as shown in Fig. 1) to generate new FR molecules with desirable overall performances in EP. Importantly, our as-developed regression and classification ML models can precisely predict the impacts of as-generated new FRs on the flame-retardant and mechanical properties, as well as T_g of EP. As-generated three FR molecules are chemically synthesized and show desired flame retardancy in EP at low loading levels ($\leq 3\text{wt}\%$). Moreover, the as-prepared flame-retardant EP coating demonstrates great potential for preventing thermal runaway propagation (TRP) in battery packs by applying it onto wood sheets. This work offers a first-of-its-kind GAI-based framework (GAI4FR) that can accelerate the creation of next-generation high-performance FRs and flame-retardant polymers, while minimizing reliance on conventional, costly, time-consuming trial-and-error methods. It also paves the way for creating other advanced functional materials for multiple high-end applications.

2. De novo flame retardant molecule generation

Generative AI (GAI) models have recently been employed for de novo molecule generation [33]. These strategies range from purely symbolic methods, e.g., genetic algorithms [34], to advanced AI-based methods, e.g., transformers [35], generative adversarial networks [36], recurrent neural networks (RNNs) [37], variational autoencoders [38], and graph neural networks [39]. In parallel, hybrid ML methods that integrate AI models with reinforcement learning (RL) in the algorithmic action space have also been explored [40]. In this work, the first task is to generate or discover new flame-retardant molecules via AI models, and RNN is employed for de novo design and generative purposes by leveraging sequential-nature data [41]. This generative model has already been successfully applied in new drug development, and thus shown great potential in discovery of new flame retardants [42]. For instance, Segler et al. [43] trained an RNN on the canonicalized Simplified Molecular Input Line Entry System (SMILES),

representing molecules, to simultaneously learn the distribution of chemical space and the syntax of the language. Yu et al. [44] used the generative network for tasks such as poem production by employing an adversarial network to estimate the expected return for state-action pairs selected from the RNN.

In this work, we propose a policy-based reinforcement learning method to fine-tune RNNs for episodic tasks, specifically for generating flame-retardant molecules with specified desired characteristics. The dataset of flame-retardant structures for epoxy resin in SMILES format is adopted from [30] and used to train an RNN. The pre-trained RNN is then optimized using reinforcement learning to achieve a specific reward. The reward is defined by a scoring function that guides the reinforcement learning to generate molecules with desired properties. Here, we also develop a scoring function to reward compounds with functional groups, such as phosphonate, hydroxy, sulfur-containing, and carbonyl groups, as these groups are highly effective in improving the flame retardancy of EP [45].

2.1. Module I: SMILE string generation for FR discovery

2.1.1. RNN long short-term memory

RNNs are neural network architectures that leverage the symmetry between steps in sequential data while retaining the most important details of previously learned steps, which can influence the interpretation of the current step [40,46]. These networks are usually composed of long short-term memory (LSTM) units or gated recurrent units, which store and transmit data from one state to another. Unlike the traditional RNNs, the LSTM model does not have vanishing or exploding issues. SMILES strings, serving as the network's input for de novo molecule generation, are one-hot encoded into binary vectors suitable for training using maximum likelihood estimation. At each training step, a probability distribution, $P(X^t)$, is generated, and the model is trained to maximize the likelihood of correctly predicting the symbol X at time t , given the previously observed symbols. This can be implemented by minimizing the negative log-likelihood, which is the cost function denoted as follows:

$$J(\theta) = - \sum_{t=0}^T \log P(x_t | x_{t-1} \dots x_0) \quad (1)$$

The cost function, $J(\theta)$, is minimized based on the network parameter, θ , and is applied to a batch, which is a subset of all training dataset. To update θ at step t , the gradient of the prediction is used for the predicted log-likelihood, and this approach is called back-propagation. After training, the LSTM model can generate new sequences based on the conditional probabilities that it has learned from the training data. In this work, we train the LSTM using SMILES representations of some flame retardants in previous work [30]. However, another challenge arises during the training procedure, because the RNNs requires a large amount of training data (at least 10^4 or 10^5 samples) to achieve well-trained results. To overcome the challenge, the data augmentation technique can be utilized to increase the data size and diversity, as demonstrated by Arús-Pous et al. [47]. Since there are only 315 flame retardants SMILES in [30], a SMILES data augmentation technique [48] is used to generate syntactically valid SMILES strings for each molecule. This process can increase the number of SMILES strings from 315 to 30000, thus facilitating the training quality.

The configuration used in this study consists of three LSTM layers with a size of 512 and no dropout, and an embedding layer with a size of 256. The configuration is trained on the augmented dataset for 5 epochs using a batch size of 128 and 8, with the ADAM optimizer and learning rate of 0.0005, which are the default values. Inspired by previous work on RL of generative chemical molecules [49], we applied the methods for training and im-

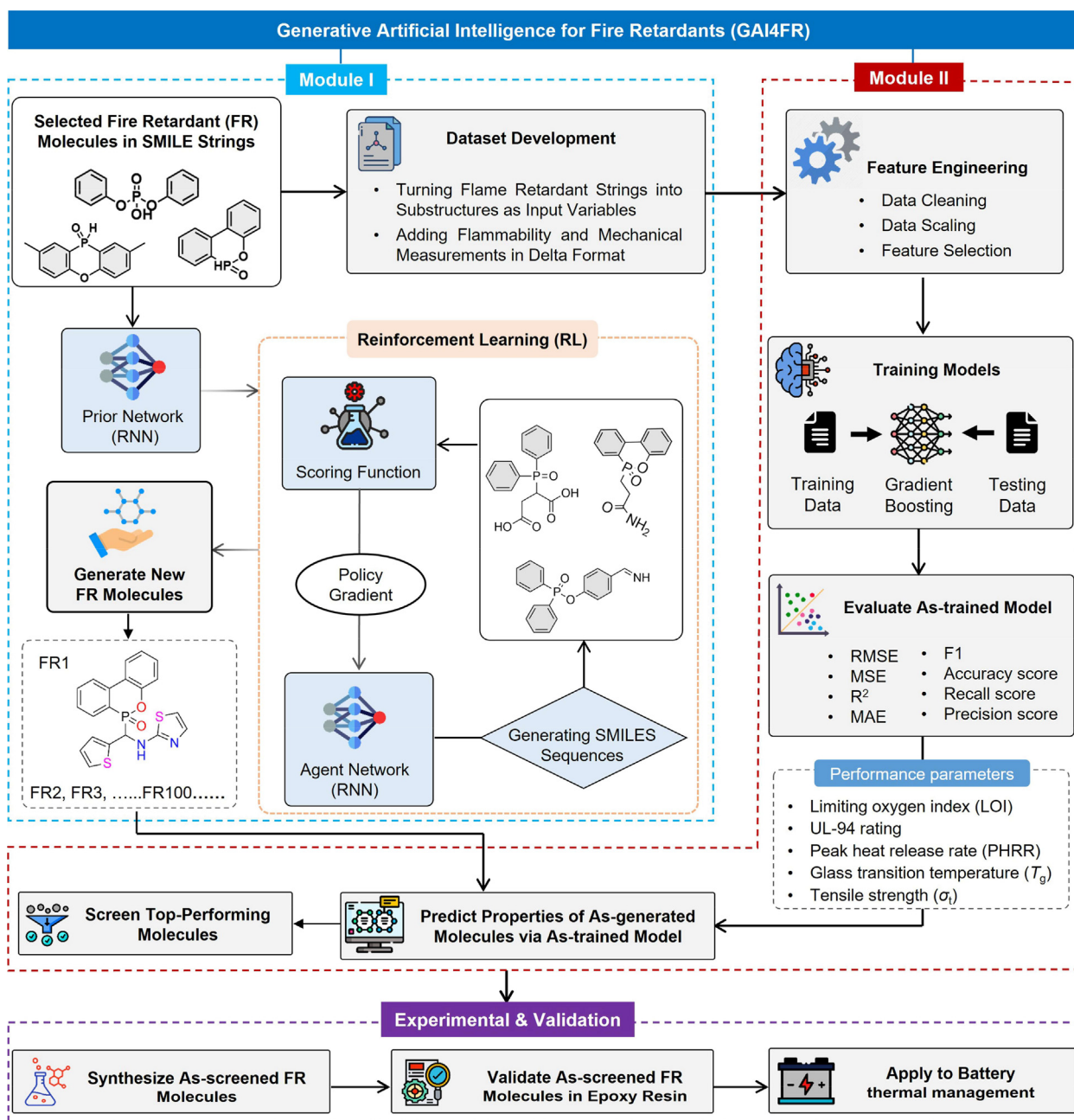


Fig. 1. Illustration of the as-proposed GAI4FR framework for the de novo design of flame retardants for epoxy resins and their screening, synthesis, validation and application.

plementing the scoring function. The training was conducted on Google Colab Pro+ with A100 GPU.

2.1.2. Reinforcement learning (RL)

As mentioned before, reinforcement learning (RL) is used to optimize the pre-trained LSTM. The aim of using RL is to guide the generation process of FR molecules towards structures that can give higher flame retardancy efficiency in EP. In a given state $s \in S$, an Agent must choose an action $a \in A(s)$, where S represents the possibilities in every state (here, the SMILES string), and $A(s)$ denotes the actions for possible states. To map a state to the probability of actions, a policy $\pi(a|s)$ is used. In reinforcement learning, problems are considered as a Markov Decision Process, where actions are chosen based solely on all information on the current state without reliance on past states [50]. The corresponding reward (r_t) indicates how appropriate the action was taken,

and the accumulated rewards from time t to final time T define the long-term return, which is $G = \sum_t^T r_t$. Based on rewards collected and actions taken, RL can optimize the agent policy to increase the expected return. The endpoint at step T is considered an episodic task, where generating SMILES serves as an example [51]. The agent can be trained based on states and actions generated by itself (on-policy method) or other means (off-policy method) [51].

Generally, there are two different approaches in RL, including value-based and policy-based RL [51]. Value-based RL focuses on learning a value function that describes the anticipated return of a given state. A policy is then set to maximize the expected value of the state. In comparison, the aim of policy-based RL is to learn a policy directly. In this work, the policy-based RL is used because it can learn an optimal stochastic policy [52]. This approach then begins with a prior sequence model, which needs to be fine-tuned

using some specific scoring functions. To establish a prior policy, an RNN should be trained on a large dataset of molecules. The prior policy is achieved by calculating the negative log-likelihood. By incorporating a prior policy into the loss function, RL explicitly regularises the optimisation process. Here, we implemented REINVENT as RL, which has been proven a highly effective method used in molecule generation [53]. As presented in Fig. 1, RL ensures that the RNN, as the agent, is optimized to retain the knowledge obtained from the previous RNN (the prior). The prior policy and scaled reward (adjusted by a scaling factor) are then combined to define an augmented likelihood, which serves as the target policy for the agent and the loss function. The loss function is updated to indicate the difference between the agent policy and target policy, as follows:

$$\log P_U(A) = \log P_{\text{prior}}(A) + \sigma R_T \quad (2)$$

$$J(\theta) = [\log P_U(A) - \log P_{\text{agent}}(A)]^2 \quad (3)$$

where $R_T \in [0, 1]$ is computed by the scoring function in a Markov Decision Process at the end of the episode [54].

$$J(\theta) = \frac{(1-\alpha)}{S} \sum [\log P_{U_{\text{prior}}}(A) - \log P_{\text{agent}}(A)]^2 + \frac{\alpha}{S} \sum [\log P_{U_{\text{best}}}(A) - \log P_{\text{agent}}(A)]^2 \quad (4)$$

A batch of molecules n with batch size S is collected from the current agent and the best agent to reinforce the selection of high-scoring molecules during optimization. Eq. (4) displays the loss-weighted average across agents, scaled by α , although the loss function for each agent remains the same as in Eq. (3).

2.2. Module II: Autonomous evaluation

In this study, we provide five databases by using data from the previous work [30] and their corresponding performances in terms of LOI, PHRR, UL-94, tensile strength (σ_t), and glass transition T_g of EP. Although datasets for LOI, PHRR, and UL-94 based on chemical descriptors were already created in the work [30], a new dataset is built based on the number of chemical substructures present in the structure of each flame retardant molecule. Hence, the input variables for each flame retardant represent the number of its specific chemical substructures. This allows us to investigate the impacts of each chemical substructure on target properties. A schematic of the data construction is provided in Fig. S1 in the Supporting Information, where two reported FR molecules with relatively high flame-retardant efficiency are adopted as samples [55,56].

Moreover, the differences in LOI, PHRR, σ_t , and T_g between the EP composite containing flame retardants and pure EP are added into the dataset, which can help reduce the data collection errors and enhance the accuracy of training. The equation for calculating the property differences between flame-retardant EP composite and pure EP is provided as follows:

$$\Delta P = P_{\text{fire-retardant EP}} - P_{\text{EP}} \quad (5)$$

where P corresponds to LOI, PHRR, σ_t , and T_g , respectively.

The LOI, UL-94, PHRR, tensile σ_t , and T_g databases consist of 916, 826, 251, 234, and 235 data points, respectively. A detailed explanation of all input variables can be found in Table S2. Based on the nature of the target variable, whether they are categorical or numerical values, it is essential to set a clear strategy for the question. In fact, regression models can be used for parameters with a numerical nature, such as T_g , tensile strength, LOI, and PHRR. In contrast, UL-94 is a graded, categorical test with a classification strategy.

In this study, we developed the ML models using Scikit-learn in Python [57]. The data are first split into training and test sets, standardized, and then the input variables are selected [57]. Afterward,

algorithms for classification and regression are trained and their performances are evaluated [58]. The experiments were conducted on a Windows 10 Education operating system, running on a system equipped with an Intel Core i7-10700 CPU and 16 GB of RAM. The AI models were implemented using Python 3.10 in a Jupyter Notebook environment provided by the University of Southern Queensland. The code and data used in this paper are provided at https://github.com/Pooya1234567/Epoxy_Flame_Retardant.git.

3. Results and discussion

3.1. Performance evaluation of machine learning models

In this section, we focus on evaluating the accuracy of AI models in predicting LOI, PHRR, UL-94, σ_t , and T_g . Generally, the coefficient of determination (R^2), mean squared error (MSE), root mean square error (RMSE) and mean absolute error (MAE) are metrics used to evaluate the performance and accuracy of trained models. Most of their predicted values closely align with the experimental ones (see Fig. 2). The model shows strong performance in predicting σ_t , achieving an R^2 of 0.99 for the training set and 0.88 for the test set, with an MSE of 33.86, a RMSE of 5.81, and an MAE of 4.17. The model also shows high accuracy for LOI, where R^2 was 0.956 for the training set and 0.759 for the test set. The MSE and RMSE are 3.48 and 1.8, respectively, indicating the high accuracy of the training.

As for both PHRR and T_g , the as-developed models show slightly low accuracy, but still high MSE and RMSE values. R^2 values of the training set are 0.972 and 0.747 for the testing set for T_g , and 0.981 for training and 0.816 for testing in the PHRR model. Regarding UL-94 ratings, we assess the training performance of models by using accuracy, precision, recall, and F1 score, with the results summarized in Table S3. The accuracy, precision, recall, and F1 score of the UL-94 model are respectively 0.905, 0.645, 0.68, 0.645, and 0.652. The reason for the relatively low performance of the UL-94 model is presumably attributed to an imbalance in the dataset, with most data concentrated in the V-0 and V-1 categories, as confirmed by the previous work [30]. Overall, our as-trained models demonstrate good performances and high accuracy, which can enable the performance evaluation of new FR molecules generated by the GAI4FR in the EP matrix.

3.2. Feature importance analysis

It is critically important to assess input variables that govern the performance metrics of flame retardants in terms of LOI, UL-94, PHRR, σ_t , and T_g . We conducted a feature importance analysis to uncover which input variables play a key role. The scale of measurement ranges from 0 to 1, with the top five most influential features shown in Fig. 3. Among the 103 substructures and addition amount as input variables, the addition amount of FRs is found to be the most impactful feature for all five variables, which is well consistent with previous findings with respect to LOI, UL-94, and PHRR [30]. The impactful substructures are also identified for each property. Both o-phthalodinitrile (S30) and phosphaphenanthrene (S58) groups exhibit the highest impact on LOI, followed by phenyl (S21) and p-hydroxyphenyl (S8) groups (Fig. 3(a)). In terms of UL-94 ratings, phosphonooxy (S65), 1,3,5-triazine-2,4,6-triamine (S28), 10-hydroxy-5,10-dihydro-10 λ -phenophosphazinin-10-one (S81), and phosphaphenanthrene (S58) are the most influential substructures. For PHRR, the substructures, including phenyl (S21), trimethyl-1,3,5-triazinane-2,4,6-trione (S51), phenylboronic acid (S44), and methoxyethane (S13), are the most influential. Regarding T_g , the substructures 4-(2-hydroxyethoxy)-4-oxobutanoic acid (S41) and ethylene oxide (S33) are the two most influential substructures, followed by phosphonooxy (S65) and 2-hydroxy-

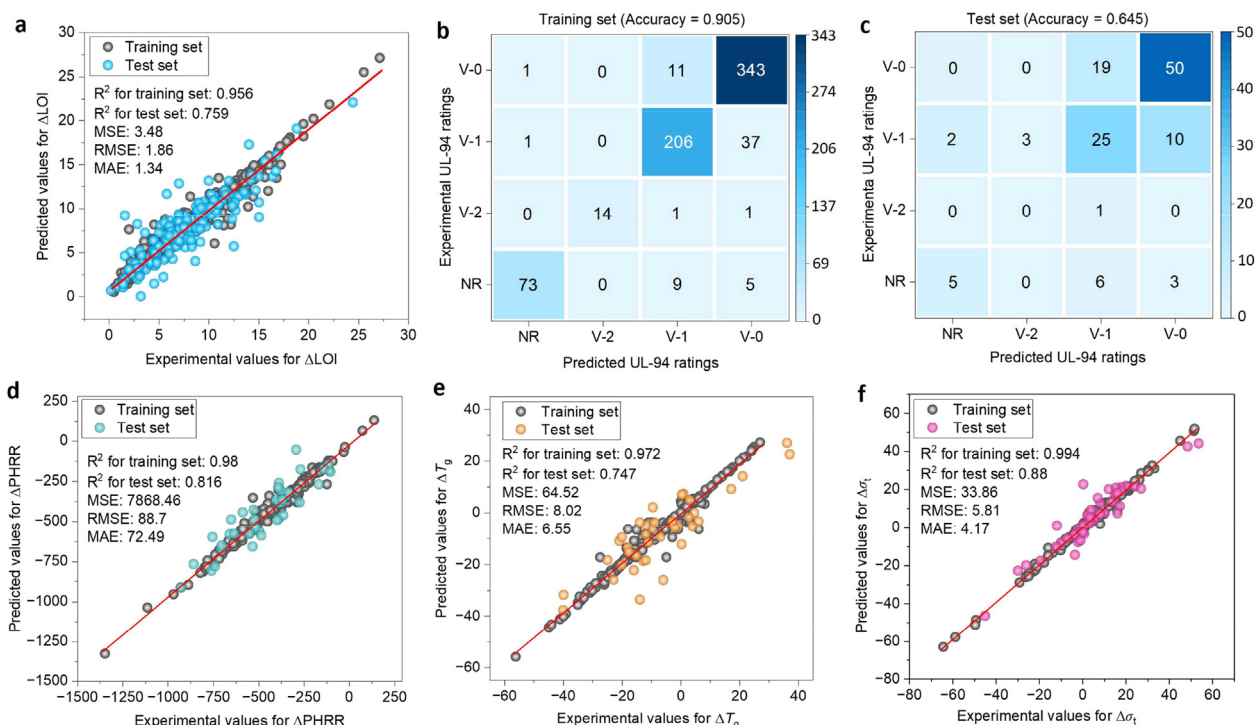


Fig. 2. Training and test sets for (a) LOI, (b) UL-94, (d) PHRR, (e) T_g , and (f) σ_t by as-trained machine learning models.

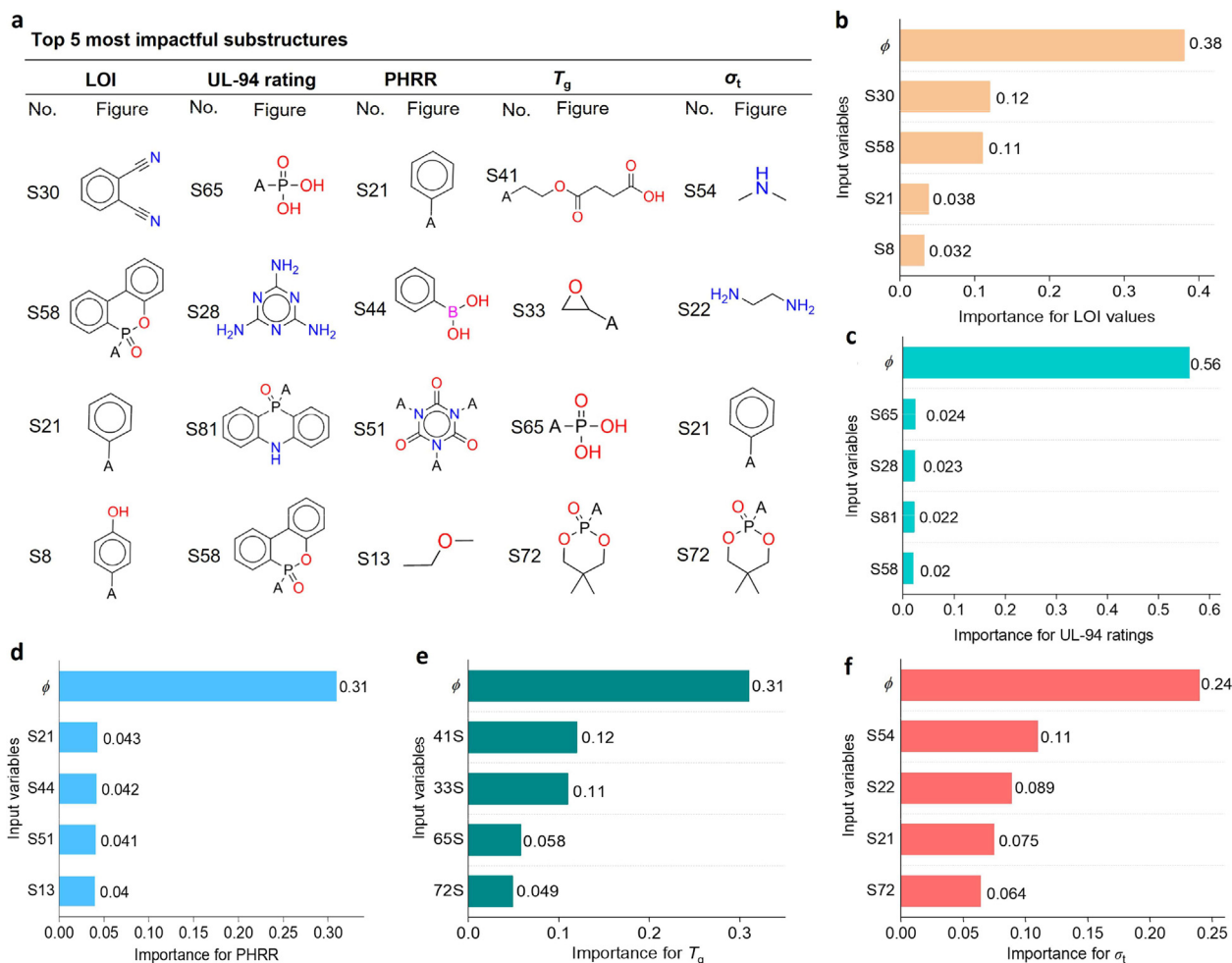


Fig. 3. Feature importance of input variables for performance parameters in machine learning models. (a) Most impactful substructures, and importance of input variables for (b) LOI, (c) UL-94 rating, (d) PHRR, (e) T_g , and (f) σ_t .

5,5-dimethyl-1,3,2λ⁵-dioxaphosphinan-2-one (S72). With respect to σ_t , dimethylamine (S54) shows the highest impact, followed by ethane-1,2-diamine (S22), phenyl (S21), and 2-hydroxy-5,5-dimethyl-1,3,2λ⁵-dioxaphosphinan-2-one (S72).

Though the feature importance analysis identifies features that have a key impact on the target property, it does not reveal whether their impact is positive or negative. To address this issue, we also performed a feature correlation analysis to assess the positive and negative effects of the features obtained in the feature importance analysis (see Table S4). Based on the results obtained for LOI, the addition amount (ϕ) and substructures (S30 and S58) show a positive correlation, indicating that the increase in their contents can enhance LOI values. In comparison, the substructures S8 and S21 exhibit a negative correlation, suggesting that the increase in their addition amount decreases LOI values. The ϕ and the substructure S58 have a positive correlation with the UL-94 rating. In contrast, the substructures S65 and S28 show a negative correlation, suggesting that their presence is likely to negatively affect the UL-94 rating. Regarding the PHRR, increasing the ϕ along with the substructures 44S, 51S, and 13S is likely to cause a reduction in PHRR since the correlation appears to be negative for these variables. Regarding σ_t , an increase in the ϕ of FRs appears to have a negative impact, but the substructures S22, S21, and S54 are likely to enhance the σ_t of the EPs due to the positive correlation. In comparison, the substructure S72 may have an adverse impact on the σ_t at high ϕ . An increase in the ϕ of FRs can reduce T_g , as indicated by the observed negative correlation, mainly because the presence of a high amount of FRs can lead to a plasticization effect on EP. Overall, the ϕ of FRs and the substructures have significant but different impacts on the five target property indices. This means that one or two target properties need to be compromised to some extent to identify some FR molecules with better overall performances in EP.

3.3. Experimental validation

After training the generative RNN model via RL on flame-retardant data in the SMILES format, a library of newly generated FR molecules are obtained. Each molecule is evaluated to identify those that are chemically valid, synthetically feasible, and demonstrate high flame-retardant efficiency based on predictions of our regression and classification ML models. We aim to identify the generated molecules that are predicted to endow EP with a UL-94 V-0 rating and a high LOI at a low ϕ (e.g., 2wt% or 3wt%). By doing this, we identify three flame retardants (P1, P2 and P3) (see Fig. 4(a)) that meet these criteria and show high performances at a low ϕ . To validate their predicted performances in EP, they are chemically synthesized, and their chemical structures are confirmed (Fig. S2). A list of other generated molecules that fail to meet the criteria is also provided in Fig. S3, and their corresponding predictions are summarized in Table S5.

Prior to the experimental validation of the overall performances of the as-generated three new flame retardants (P1, P2 and P3) in EP, including LOI, UL-94 rating, PHRR, σ_t , and T_g , we first use our trained models to predict their properties. All the predicted performance parameters by our model (this work) and the previous work [30] are then compared with the experimental values to assess the accuracy of the two models in predicting target performances. Since the predictions of our models are based on Eq. (5), we estimate the properties of flame-retardant EPs by adding the model-predicted difference to the experimental value of pure EP.

For LOI values, our predicted LOI values are very close to the experimental ones, and our as-trained model shows higher accuracy than the previous work [30] (Fig. 4(b)). For example, our predicted LOI value for flame-retardant EP with 1wt% of P1 only shows a 0.8% discrepancy with respect to the experimental value,

whereas a discrepancy of nearly 4.0% for the other model [30]. Regarding UL-94 ratings, both our model and the previous model show precise predictions as compared to the experimental values (Table S6). In addition, our model can accurately predict the UL-94-rating for both P2 and P3 molecules. The experimental results for both LOI and UL-94 rating parameters clearly validate the high accuracy of our trained model. With respect to the PHRR values for flame-retardant EPs containing as-synthesized P1, P2 and P3, our as-trained model demonstrates far higher prediction accuracy than the previous model [30] as compared with the corresponding experimental values (Fig. 4(d) and Table S7). For P1-containing EPs, our predicted PHRR values are very close to the experimental values. For instance, there is a difference of only 1.62% between the experimental (Exp.) and predicted (Pred.) PHRR values for the flame-retardant EP containing 1 wt% of P1. Regarding the EP samples containing P2 and P3, our model can also predict their PHRR values accurately.

In addition, the predicted T_g values of P1-containing EP systems are close to the experimental ones (Fig. 4(e)), further verifying the high accuracy of our model. Furthermore, our model shows an ability to accurately predict the σ_t of flame-retardant EP systems at FR contents up to 1wt% (Table S8 and Fig. 4(f)). The predicted σ_t values of EP containing 1wt% of P1, P2, and P3 approach their experimental ones, and a similar result can be found for EP with 2wt% of P1. When the FR content ranges from 3wt%–7wt%, the predicted σ_t deviates considerably from the experimental value, indicating that the reduced predictive accuracy persists across different loadings (3wt%–7wt%). This finding confirms that the discrepancy arises from data scarcity rather than a fundamental limitation of the model in handling higher flame-retardant concentrations. This is understandable because an inadequate amount of data available can introduce limited knowledge to the ML algorithm during training, thus reducing the prediction accuracy of the as-trained models to a certain degree.

Notably, our as-synthesized flame retardants (P1, P2, and P3) generated by our model demonstrate higher flame-retardant efficiency compared to previously reported FRs [59–68] in terms of LOI values and UL-94 ratings (Fig. 4(g)). Specifically, they can impart higher LOI values along with the highest UL-94 V-0 rating to EP even at relatively low loading levels, while enabling the final EP to achieve higher T_g values. These results strongly validate the high accuracy and effectiveness of our model in generating highly efficient flame retardants for EP. Also, it further underscores our model's promising potential for practical applications while minimizing costly trial-and-error experiments.

It should be noted that the predictive accuracy of the present ML models is highest within the flame-retardant loading domain, where the model was well-trained on sufficient training data. As for a high flame-retardant loading regime, the predicted values show less accuracy, as the limited number of high-loading data has not helped the model to reach the highest accuracy. Therefore, the current model is primarily optimized for low-loading levels, mostly applicable for highly efficient flame retardants. Expanding the training dataset to include more samples with higher FR loadings would likely enhance predictive accuracy in this regime.

3.4. Real-world applications for battery thermal runaway control

EP resins have been widely applied as coatings for different substrates, including wood, steel and concrete, due to universal interface adhesion and exceptional chemical resistance. Because the as-generated P1 flame retardant has demonstrated the highest flame retardance in EP, we then use it to prepare flame-retardant EP coatings for flammable timber. The EP coating with 3wt% of P1 (EP/3%P1) can endow the flammable wood with exceptional flame retardancy. Specifically, a 500 μm thick EP/3%P1 coating enables

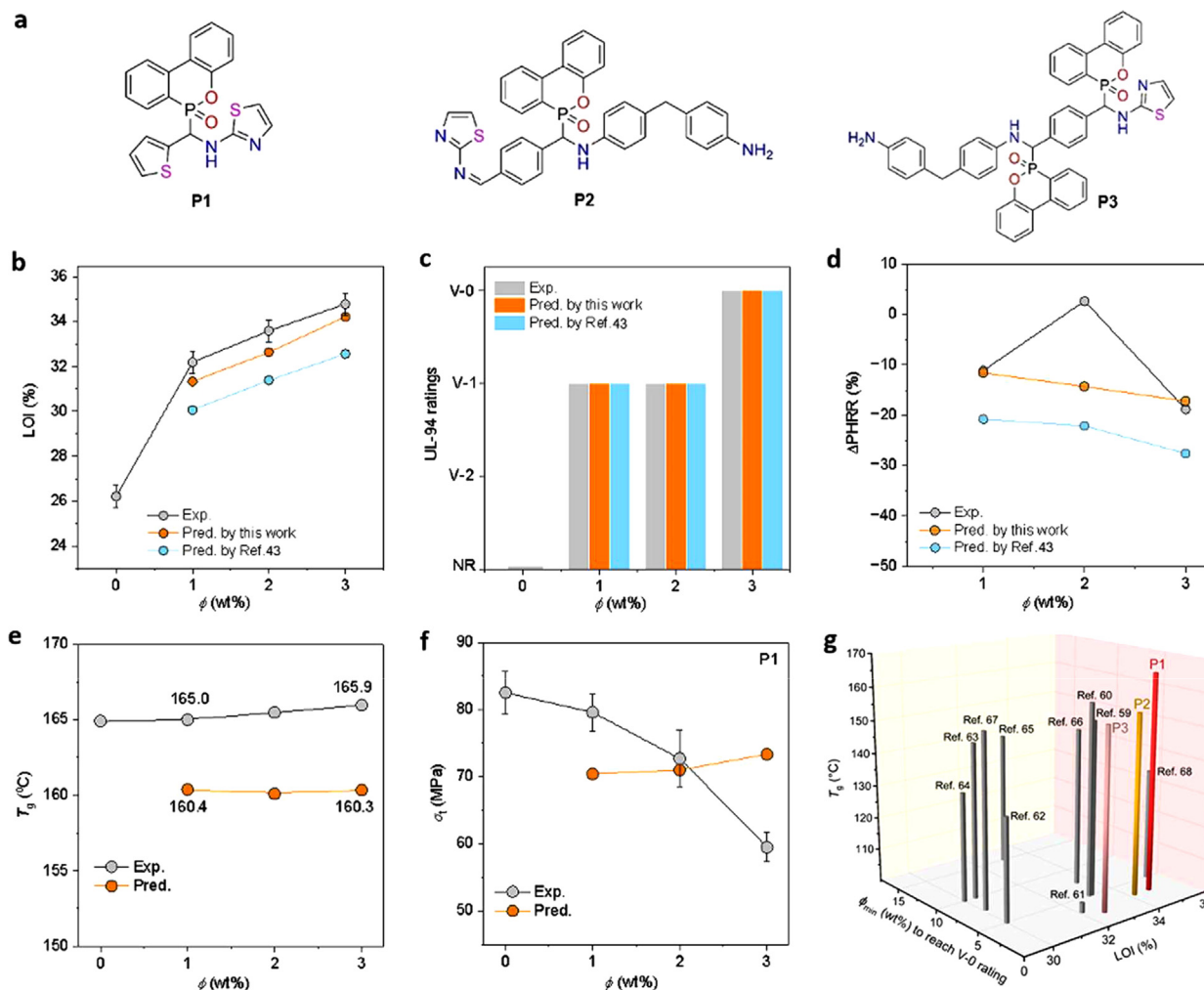


Fig. 4. (a) Chemical structures of as-predicted three flame retardants (P1, P2, and P3) for EP, and impacts of P1 flame retardant on fire performances of EP: (b) LOI, (c) UL-94 rating, and (d) Δ PHRR%, and on (e) T_g and (f) σ_t of EP. (g) Comparison between flame-retardant EP with 3.0wt% P1 and reported flame-retardant EP counterparts [59–68] in terms of the ϕ_{min} to reach V-0 rating, T_g and LOI values.

the wood to achieve a high LOI value of 28.5%, a desired UL-94 V-0 rating, an increased time to ignition (t_{ign}) of 29 s, and a high char residue of 14.9% (Fig. 5(a and b)). Such an extraordinary flame-retardant efficiency of the as-prepared EP coating is primarily because of its ability to enhance the anti-ignition and carbonization ability of the wood substrate.

We further demonstrate the potential application of as-prepared sustainable flame-retardant wood as a heat shield for preventing the thermal runaway propagation of lithium-ion batteries (LIBs). LIBs are being commonly applied in electric vehicles (EVs) and energy storage systems due to their long cycle life and high energy density, but their widespread use has continuously caused significant fire disasters. Due to the inherent electrochemical reaction nature, LIBs are often susceptible to thermal runaway (TR) when exposed to external stimuli, such as mechanical forces, heat sources, and electricity. The TR of one battery cell can trigger the chain TR of the entire battery pack, which eventually leads to safety accidents, e.g., fire and even explosion. Therefore, it is essential to use heat-insulating and flame-retardant materials as heat shields between cells to prevent the thermal runaway propagation and to avoid catastrophic fire accidents. Obviously, LOI, UL-94, and PHRR are the key indicators to assess the intrinsic flame-retardant performance of EPs. Higher LOI values, UL-94 V-0 ratings, and lower PHRR collectively indicate stronger flame inhibition and

reduced heat release, which are directly beneficial for mitigating heat transfer and suppressing thermal runaway propagation in battery systems.

Accordingly, the as-prepared EP/3%P1-coated wood can serve as an ideal heat shield for LIBs due to its sustainability, exceptional flame retardancy (high LOI, UL-94 V-0 rating, and low PHRR), and low thermal conductivity. To evaluate its effectiveness, a TRP test was conducted on a battery pack comprising two LIB cells (rated capacity: 2 or 40 Ah) with or without various heat shields, including commercial SiO_2 aerogel felt, commercial flame-retardant EP-coated wood, and EP/3%P1-coated wood, using a home-made set-up (Fig. 5(c)). Without any heat shield, both battery Cells 1 and 2 (rated capacity: 2 Ah) experience rapid TR once Cell 1 is heated by a high-power copper heater, with Cell 2 failing within only 137 s (Fig. S4(a)). When commercial flame-retardant EP-coated wood, 3 mm-thick pure EP or wood sheets are used as heat shields, none can prevent the TR of the battery pack during testing (Figs. 5(d, d₁), S5 and S6). In contrast, both commercial SiO_2 aerogel felt and EP/3%P1-coated wood effectively suppress the TR of battery Cell 2 under both rated capacities (2 and 40 Ah), as presented in Figs. 5(e–h), S7 and S9. When the P1 content increases to 5wt% or 7wt%, the resultant EP/5%P1 and EP/7%P1 coated woods can completely prevent TR propagation. Similarly, P2- and P3-containing EP-coated woods (flame retardant content: 3wt%, 5wt%, or 7wt%)

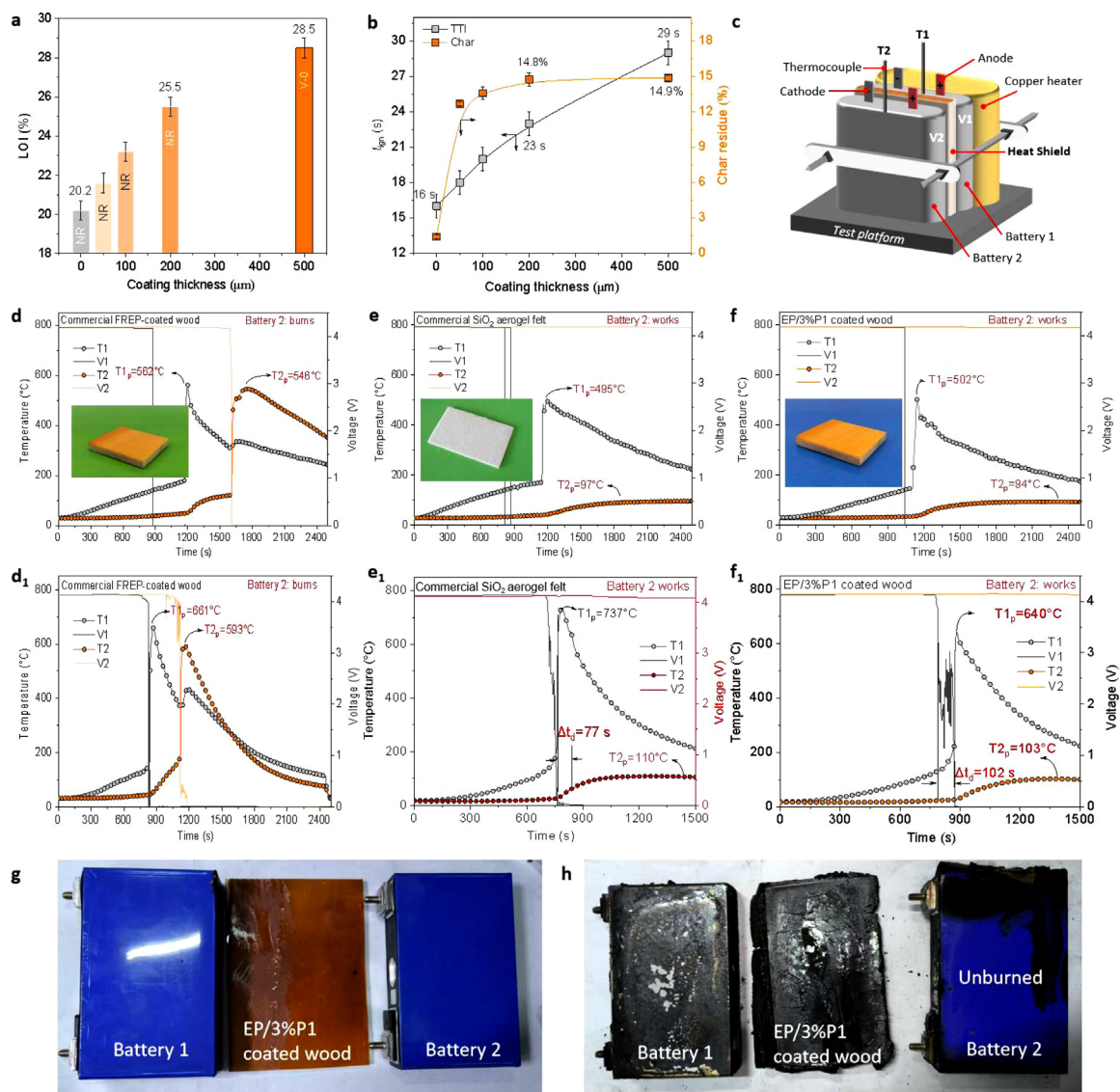


Fig. 5. (a) LOI values and UL-94 ratings and (b) time to ignition (t_{ign}) and char residues of wood coated with varied EP/3%P1 coating thickness. (c) Schematic illustration for the home-made set-up for thermal runaway propagation testing. Backside temperatures (T1 and T2) and battery voltages (V1 and V2) of two batteries protected by a heat shield: (d, d₁) commercial flame-retardant EP coated wood, (e, e₁) commercial SiO₂ aerogel felt, and (f, f₁) EP/3%P1-coated wood (coating thickness: 200 μm, (d–f): LIB rated capacity of 40 Ah, and (d₁–f₁): LIB rated capacity of 2 Ah). Digital images of two batteries protected by EP/3%P1-coated wood (g) before and (h) after TR testing.

also show excellent TR suppression performance (Figs. S4 and S8). Notably, despite the much lower thermal conductivity of commercial SiO₂ aerogel felt (~ 0.022 W/(m K)), the EP/3%P1-coated wood ($\lambda = 0.192$ W/(m K)) provides superior thermal protection, as evidenced by a slower temperature rise and lower final temperature of Cell 2. Furthermore, SiO₂ aerogel felts can pose respiratory risks during use and disposal, whereas the EP/3%P1-coated wood heat shield is significantly more sustainable and user-friendly, being derived from renewable timber with only a thin layer of flame-retardant coating (coating thickness: 500 μm). In addition, compared with other commercial heat shields, the as-prepared EP/3%P1-, EP/3%P2- and EP/3%P3-coated woods exhibit significantly lower costs, demonstrating their excellent cost-effectiveness (Table S9). The prices are based on 3 mm thickness, and the labour and manufacturing costs are already included for EP/3%P1-, EP/3%P2- and EP/3%P3-coated woods. The superior heat protection of as-developed EP/3%P1-coated wood is attributed to its anti-ignition, strong charring ability, superior flame retardance, and low λ (Table S10). These results confirm that EP/3%P1-coated wood can ef-

fectively prevents the thermal runaway propagation of the battery pack as a heat shield.

4. Conclusions

In summary, we have developed a groundbreaking de novo design approach based on the GAI4FR framework to develop highly efficient flame retardants for EP. This process utilizes RNNs combined with RL to design flame-retardant molecules. Meanwhile, ML classification and regression models are applied to predict the impacts of as-generated new FR molecules on key flame-retardant and mechanical properties of EP, including LOI, UL-94 rating, PHRR, T_g , and σ_t , enabling us to screen and identify highly effective FRs with better overall performances in EP. Our trained model has successfully identified three FR molecules (P1, P2 and P3) that meet these criteria, and the experimental results fully validate the accuracy of the prediction of our AI models. The newly generated P1 flame retardant shows excellent flame retardancy in EP as expected, and the resultant EP/3%P1 flame-retardant coating can sig-

nificantly enhance the flame retardancy of wood. We demonstrate that as-developed EP/3%P1-coated wood can serve as a sustainable heat shield material for preventing thermal runaway propagation of battery packs, which facilitates minimizing the possibility of catastrophic fire accidents when one battery cell experiences thermal runaway. This work offers a first-of-its-kind GAI4FR framework for the discovery of next-generation high-performance flame retardants and lays a solid foundation for the accelerated creation of flame retardants and other functional materials for multiple industrial applications, such as building & construction, energy storage, electronic devices and beyond.

5. Experimental section

This section is included in the Supporting Information.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Pooya Jafari: Writing – original draft, Data curation, Conceptualization. **Guofeng Ye:** Formal analysis, Data curation. **Junling Wang:** Formal analysis, Data curation. **Tao Chu:** Writing – original draft. **Boyao Hou:** Formal analysis. **Di Wu:** Writing – review & editing. **Siqi Huo:** Writing – review & editing. **Hao Wang:** Writing – review & editing. **Ziqi Sun:** Writing – review & editing. **Pingan Song:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jmst.2026.03.044](https://doi.org/10.1016/j.jmst.2026.03.044).

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